Arithmetic & Number Sense

Order of operations

1 Evaluate $12 + 6 \times (9 - 5) \div 3$.

2 Evaluate $8 + 2 \times 5^2 - 30 \div 6$.

3 Evaluate $\frac{18 - 3 \times 4}{2} + 7$.

4 Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.

Powers and roots

5 Evaluate 3^4.

6 Evaluate $\sqrt{144}$.

7 Evaluate $2^3 \times 2^2$ using the laws of exponents.

8 Evaluate $\sqrt{81} + \sqrt{16}$.

Working with fractions

9 Compute $\frac{3}{4} + \frac{5}{6}$.

10 Compute $\frac{7}{8} - \frac{1}{3}$.

11 Compute $\frac{2}{3} \times \frac{9}{10}$.

12 Compute $\frac{5}{6} \div \frac{5}{9}$.

13 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

Decimals

14 Compute $3.45 + 2.7$.

15 Compute $8.6 - 3.25$.

16 Compute 1.2×0.5.

17 Compute $7.5 \div 0.25$.

Place value and order of magnitude

18 Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \leq a < 10$.

19 Which is greater, 6×10^4 or $58{,}000$?

20 Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

Mixed numbers and improper fractions

21 Convert $4\frac{2}{5}$ to an improper fraction.

22 Convert $\frac{47}{6}$ to a mixed number.

23 Compute $2\frac{1}{3} + 1\frac{3}{4}$.

24 Compute $5\frac{1}{2} - 2\frac{2}{3}$.

Absolute value and signed numbers

25 Evaluate $|-7| + |-3|$.

26 Evaluate $-8 + 5 \times (-2)$.

27 Evaluate $|4 - 9| - |2 - 6|$.

Factors, multiples, GCD and LCM

28 Find the greatest common divisor (GCD) of 48 and 60.

29 Find the least common multiple (LCM) of 8 and 12.

30 List all the factors of 36, and state how many there are.

31 Is 91 a prime number? Justify your answer.

32 Find the greatest common factor of 84 and 126.

Divisibility rules

33 Without dividing, determine whether $4{,}578$ is divisible by 3.

34 Is $2{,}340$ divisible by 9?

35 Is 715 divisible by 5?

36 Is $1{,}244$ divisible by 4?

Estimation and rounding

37 Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.

38 Round 2.4863 to two decimal places.

39 Estimate $\sqrt{50}$ to the nearest whole number.

40 A shipment weighs approximately $2{,}850$ kg. Round this to the nearest hundred kilograms.

Number properties

41 If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?

42 Find the remainder when 47 is divided by 6.

43 Is 637 divisible by 7? Justify your answer.

44 List all prime numbers between 30 and 50.

45 If the sum of two integers is odd, is their product always even, always odd, or can it be either?

Word problems

46 A school orders 158 notebooks to be shared as equally as possible among 12 classrooms. How many notebooks does each classroom receive, and how many are left over?

47 A caterer must seat 250 guests at tables that each hold 8 people. What is the minimum number of tables required?

48 Three friends share a restaurant bill of 186 riyals equally, then each adds a 10-riyal tip. How much does each person pay in total?

49 A red light flashes every 8 seconds and a green light flashes every 12 seconds. If both flash together right now, after how many seconds will they next flash together at the same time?

50 A water tank holds 480 liters and is filled by a pump running at 15 liters per minute. How long does it take to fill the tank from empty?

51 A charity collects 2,340 riyals from 45 equal donations. How much was each donation?

MCQ — Order of operations

52 Evaluate $20 - 4 \times 3 + 6 \div 2$.

- **a.** 11
- **b.** 27
- **c.** 9
- **d.** 17

53 Evaluate $(5+3)^2 - 2 \times 7$.

- **a.** 78
- **b.** 36
- **c.** 50
- **d.** 46

54 Evaluate $\frac{45}{3+6} \times 2$.

a. 10

b. 90

c. 7

d. 15

55 Evaluate $3 + 2 \times (8 - 5)^2$.

a. 21

b. 45

c. 17

d. 33

56 Evaluate $100 - 3 \times (4 + 2 \times 3)$.

a. 64

b. 91

c. 88

d. 70

MCQ — Exponents and roots

57 Evaluate 2^5.

a. 16

b. 25

c. 32

d. 10

58 Evaluate $\sqrt{196}$.

a. 98

b. 13

c. 14

d. 15

59 Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

c. 729

d. 27

60 Evaluate $\sqrt{225} + \sqrt{64}$.

a. 17

b. 30

c. 23

d. 19

61 Evaluate $5^3 - 4^2$.

a. 109

b. 121

c. 93

d. 45

MCQ — Fraction operations

62 Compute $\frac{2}{5} + \frac{1}{4}$.

a. $\frac{1}{2}$

b. $\frac{11}{20}$

c. $\frac{13}{20}$

d. $\frac{1}{3}$

63 Compute $\frac{5}{6} - \frac{1}{3}$.

a. $\frac{1}{2}$

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{2}{3}$

64 Compute $\frac{3}{4} \times \frac{2}{9}$.

a. $\frac{5}{36}$

b. $\frac{1}{6}$

c. $\frac{2}{3}$

d. $\frac{5}{13}$

65 Compute $\frac{7}{8} \div \frac{7}{4}$.

a. $\frac{1}{2}$

b. $\frac{49}{32}$

c. $\frac{7}{2}$

d. $\frac{1}{4}$

66 Compute $\frac{1}{3}+\frac{1}{4}+\frac{1}{6}$.

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{3}{13}$

d. $\frac{3}{4}$

67 Compute $\frac{5}{9}-\frac{1}{6}$.

a. $\frac{4}{15}$

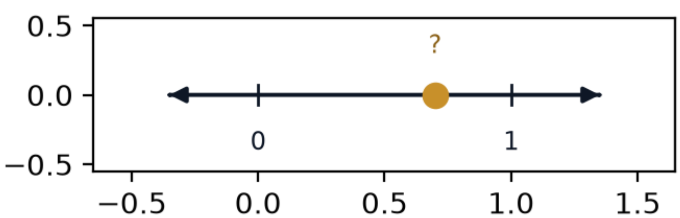
b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{7}{18}$

MCQ — Comparing fractions and decimals

68 Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?



a. The marked point (0.7)

b. $\frac{3}{5}$

c. They are all equal

d. 0.55

69 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

a. They are equal

b. Cannot be determined

c. $\frac{5}{7}$

d. $\frac{4}{5}$

70 Order from least to greatest: 0.62, $\frac{5}{8}$, 0.6.

a. $0.62 < 0.6 < \frac{5}{8}$

b. $0.6 < \frac{5}{8} < 0.62$

c. $\frac{5}{8} < 0.6 < 0.62$

d. $0.6 < 0.62 < \frac{5}{8}$

MCQ — Decimal operations

71 Compute $4.75 + 2.6$.

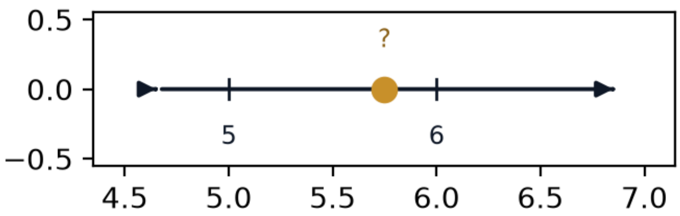
a. 7.01

b. 7.41

c. 7.35

d. 6.35

72 The number line marks the value of $9.2 - 3.45$. Where does it land?



a. 5.75

b. 6.25

c. 5.85

d. 5.65

73 Compute 1.4×0.6.

a. 8.4

b. 0.24

c. 0.84

d. 0.74

74 Compute $9.6 \div 0.4$.

a. 2.4

b. 38.4

c. 24

d. 0.24

MCQ — Scientific notation & place value

75 Write $6{,}200{,}000$ in the form $a \times 10^n$ with $1 \le a < 10$.

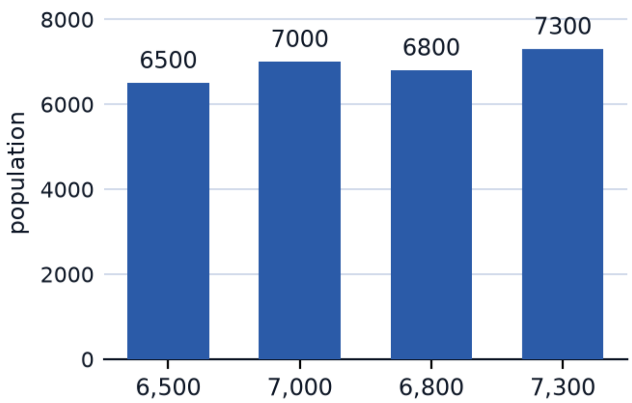
a. 6.2×10^5

b. 62×10^5

c. 6.2×10^7

d. 6.2×10^6

76 The bar chart shows four city populations. Which one equals 7×10^3?



a. 7,000

b. 6,800

c. 6,500

d. 7,300

77 Evaluate 3.4×10^4 as an ordinary number.

a. 3,400,000

b. 3,400

c. 340,000

d. 34,000

78 In the number $528{,}437$, what is the place value of the digit 8?

a. Hundred thousands

b. Thousands

c. Ten thousands

d. Hundreds

MCQ — Mixed numbers & improper fractions

79 Convert $\frac{29}{6}$ to a mixed number.

a. $4\frac{5}{6}$

b. $5\frac{5}{6}$

c. $5\frac{4}{5}$

d. $3\frac{5}{6}$

80 Convert $5\frac{3}{7}$ to an improper fraction.

a. $2\frac{1}{7}$

b. $3\frac{4}{5}$

c. $5\frac{3}{7}$

d. 5

81 Compute $3\frac{1}{4} + 1\frac{5}{6}$.

a. $5\frac{1}{12}$

b. 5

c. $\frac{4}{5}$

d. $4\frac{11}{12}$

82 Compute $6\frac{1}{5} - 2\frac{3}{4}$.

a. $3\frac{19}{20}$

b. 4

c. $1\frac{17}{20}$

d. $3\frac{9}{20}$

MCQ — Absolute value & signed numbers

83 Evaluate $|-9| + |-4|$.

a. 5

b. -13

c. 13

d. 36

84 Evaluate $-6 + 4 \times (-3)$.

a. -18

b. -30

c. 6

d. -2

85 Evaluate $|5 - 11| - |3 - 8|$.

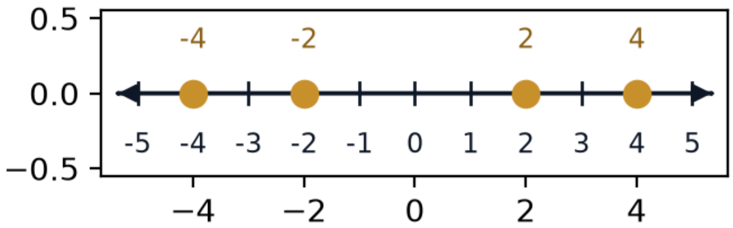
a. 11

b. 3

c. -1

d. 1

86 Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?



a. -2 and 2

b. -4 and 4

c. -2 and 4

d. -4 and 2

MCQ — GCD, LCM & factors

87 Find the GCD of 54 and 72.

a. 9

b. 6

c. 18

d. 36

88 Find the LCM of 9 and 15.

a. 3

b. 24

c. 45

d. 135

89 How many factors does 60 have?

a. 6

b. 8

c. 12

d. 10

90 Is 119 a prime number?

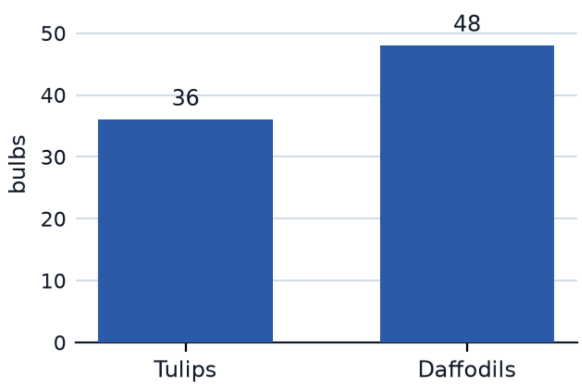
a. No — $119 = 7 \times 17$

b. Yes, it is prime

c. No — $119 = 11 \times 9$

d. Cannot be determined

91 The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?



a. 4

b. 24

c. 6

d. 12

MCQ — Divisibility rules

92 Without dividing, is $5{,}481$ divisible by 3?

a. Cannot tell without dividing

b. Yes

c. No

d. Only by 9, not 3

93 Is $2{,}116$ divisible by 4?

a. Only by 8

b. Yes

c. No

d. Only by 2

94 Is 945 divisible by 9?

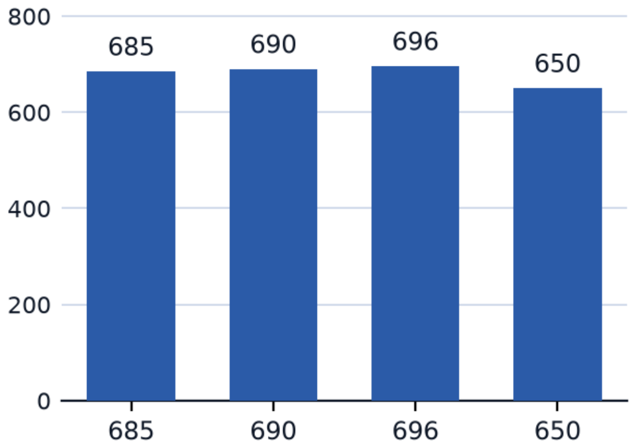
a. Yes

b. Only by 3

c. Only by 5

d. No

95 The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?



a. 690

b. 650

c. 685

d. 696

MCQ — Estimation & rounding

96 Round 147.362 to the nearest tenth.

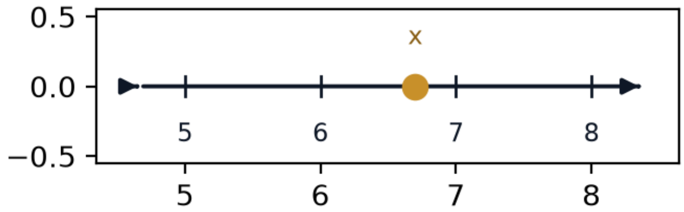
a. 147.3

b. 150

c. 147.36

d. 147.4

97 The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?



a. 6

b. 6.5

c. 8

d. 7

98 Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. Estimate $310 \times 50 = 15{,}500$, close to the exact value

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MCQ — Number properties

99 If n is an odd integer, is $n^2 + n$ always even, always odd, or can it be either?

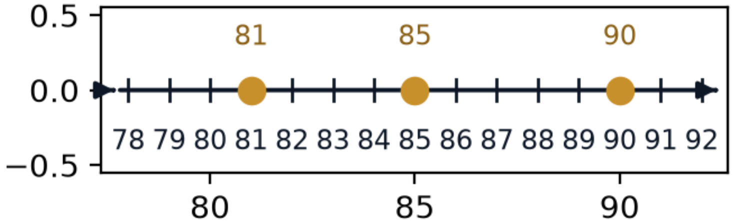
a. Can be either

b. Depends on the sign of n

c. Always odd

d. Always even

100 The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.



a. 3

b. 4

c. 5

d. 76

101 List all prime numbers between 50 and 70.

a. $51, 57, 63, 69$

b. $53, 59, 61, 67$

c. $59, 61, 63, 67$

d. $53, 57, 61, 67$